CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipelining Processing

Lecture No.- 1



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Recap of Previous Lecture



Topic Multiple Sector Access

Topic Disk Cylinder

Topic Disk Addressing

Topics to be Covered



- **Topic** Pipeline Processing
- **Topic** Pipeline Cycle Time
- **Topic** Speed Up



Topic : Parallel Processing

Parallel Processing: Simultaneous data processing

Types:

- Vector Processing
- Array Processing
- Pipeline Processing (Pipelining)



Topic : Pipeline Processing

- Pipelining is useful, When same processing is applied over multiple inputs

	cut	stitch	finish	Pack
10 min →	S1			
20 min →	S2	S1		
30 min →	S3	S2	S1	
40 min →	S4	S3	S2	S1
50 min →	S5	S4	S3	S2
60 min →	S6	S5	S4	S3
70 min →	—	S6	S5	S4
	—	—	S6	S5
				S6

90 min



Topic : Pipeline Processing

- Technique to decompose a sequential process into sub-operations
- Sub-operations are performed in segments (stages)
- Each segment perform it's respective suboperation in parallel
- **Task:** One operation performed in all segments



Topic : Pipelining Example

non-pipeline :-

if 1 shirt ⇒ 40 min

6 shirts ⇒ 6 * 40 = 240 min

pipeline :-

6 shirts ⇒ 90 min

no. of inputs	non-pipeline	pipeline
n = 6	6 * 40 = 240 min	90 min
= 7	7 * 40 = 280 min	100 min
= 8	8 * 40 = 320 min	110 min
= 9	9 * 40 = 360 min	120 min



Topic : Example

$n = 6$
$k = 4$

$4 + 6 - 1 = 9$

Clock pulse number	Segment 1 cut	Segment 2 stitch	Segment 3 finish	Segment 4 pack
1	S1			
2	S2	S1		
3	S3	S2	S1	
4	S4	S3	S2	S1
5	S5	S4	S3	S2
6	S6	S5	S4	S3
7		S6	S5	S4
8			S6	S5
9				S6



Topic : Pipeline Cycle Time

min. amount of time in which all segments can finish their respective subopcrations.





Topic : General Consideration About Pipeline

- Consider a k segment pipeline with clock cycle time = t_p to perform n tasks

no. of inputs

	no. of cycles	time
Time needed to perform 1st task	= k	$k * t_p$
—11— remaining (n-1) tasks	= n-1	$(n-1)\, t_p$
—11— all n tasks	= k+n-1	$(k+n-1)\, t_p$



Topic : General Consideration About Pipeline

- Consider a non-pipeline system that takes t_n time to perform a task

Time needed to perform all n tasks = $n * t_n$



Topic : General Consideration About Pipeline

- Performance of a pipeline is given by Speed up ratio.

$$\text{speed up} \quad (S) = \frac{n * t_n}{(k+n-1)t_p}$$

as the number of tasks increases

$n >> k$

$$S = \frac{n * t_n}{n * t_p}$$

ignore $(k-1)$

⇓

per input 1 cycle taken

For n inputs $\Rightarrow n$ cycles

$$S_{ideal} = \frac{t_n}{t_p}$$

↑ max speed up

Special Case :-

If time taken by pipeline and non-pipeline for one input are same.

then $t_n = k * t_p$

$$S_{ideal} = k$$

[NAT]

#Q. A non-pipeline system takes 50 ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10ns.

1. Determine the speedup ratio of the pipeline for 100 tasks.
2. What is the maximum speedup that can be achieved?

$t_n = 50ns$

$k = 6$

$t_p = 10ns$

$n = 100$

1. $S = \frac{100 * 50}{(6+100-1)\ 10} = 4.76$

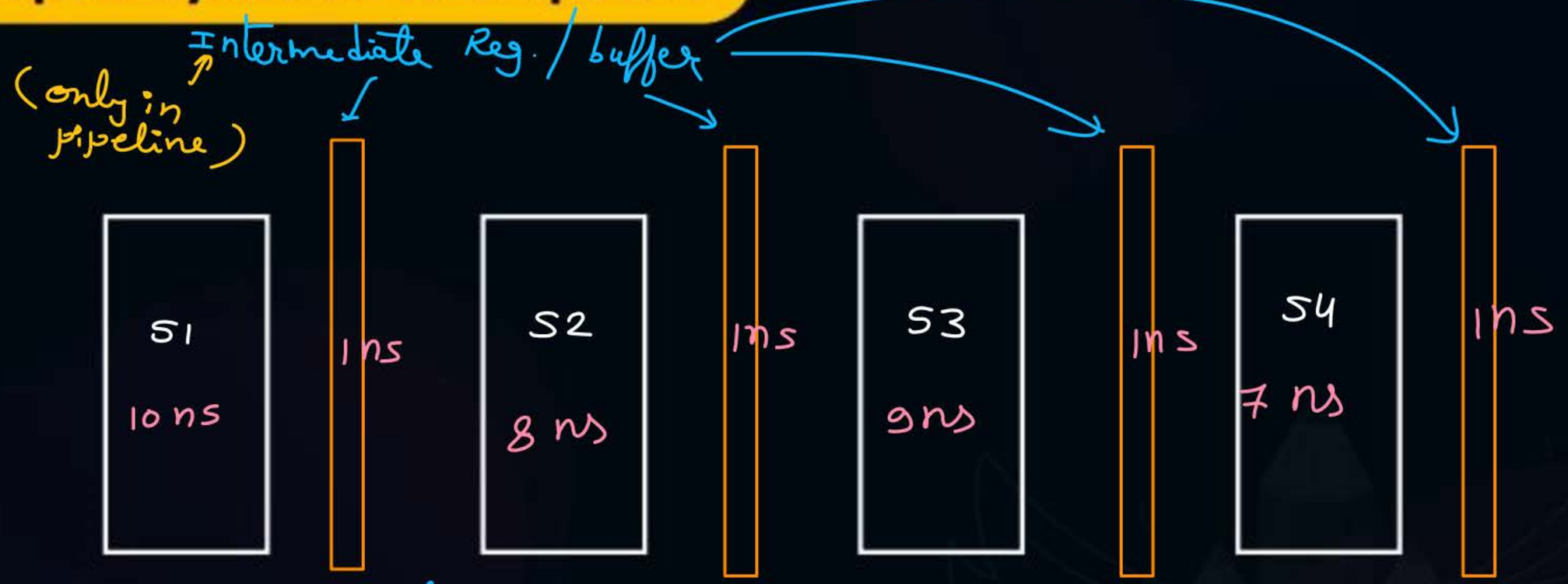
2. $S_{ideal} = \frac{50}{10} = 5$

Question :-

- → ideal condition
- → max speed up
- → value of n not given

$$S = \frac{t_n}{t_p}$$

Topic : Synchronous Pipeline



t_p = max of all segment delays = max (10, 8, 9, 7) = 10 ns

t_p = ——11—— + Reg. delay = max (10, 8, 9, 7) + 1 = 11 ns



Topic : Cycle Time in Synchronous Pipeline

t_n = sum of all segment delays = $10 + 8 + 9 + 7 = 34$ ns

If all segments will take equal delay and there is no any buffer used in pipeline, then

$$t_n = k * t_p$$

[NAT]

#Q. Consider 6 segment pipeline with segment delay of segments as 20ns, 26ns, 21ns, 21ns, 24ns and 28ns respectively. Calculate processing time of pipeline for 1000 tasks?

$k = 6$

$t_p = \max(20, 26, 21, 21, 24, 28) = 28\,ns$

$n = 1000$

$$\text{pipeline time} = (6 + 1000 - 1)\,28$$

$$= 28140\,ns$$

$$= 28.14\,\mu sec$$

[NAT]

#Q. The time delay of the four segments in pipeline are as shown follows:
$t_1 = 50$ ns, $t_2 = 30$ ns, $t_3 = 95$ ns, and $t_4 = 45$ ns.
The interface registers delay time $t_r = 5$ ns.
How long would it take to process 100 tasks in the pipeline?

$t_p = \max(50, 30, 95, 45) + 5 = 100$ ns

$n = 100$

$k = 4$

$= (4 + 100 - 1)\ 100$

$= 10300$ ns

$t_n = 50 + 30 + 95 + 45$

$= 220$ ns

$S = \dfrac{100 * 220}{10300}$

$= 2.14$

[NAT]

#Q. How can we reduce the total time about the one-half of the time calculated in above question?

Divide seg. 3 into 2 segments of delays 50ns, 45 ns

new pipeline:-

S1 ⇒ 50 ns
S2 ⇒ 30 ns
S31 ⇒ 50 ns
S32 ⇒ 45 ns
S4 ⇒ 45 ns

→ k = 5

$t_p = \max(50, 30, 50, 45, 45) + 5$
$= 55$ ns

pipeline time $= (5 + 100 - 1)\ 55$
$= 5720$ ns

Ques) In prev. question speed up achieved by new pipeline as compared to old pipeline ?

Soln

$$= \frac{10300}{5720}$$

$$= 1.8$$

Ques) If pipeline cycle time = 0.5 ns

pipeline clock rate = 2 GHz ?

Soln

$$\text{pipeline clock rate} = \frac{1}{0.5\,ns}$$

$$= 2\ GHz$$ Ans

[NAT]

#Q. Consider a non-pipelined processor with a clock rate of 5 gigahertz and average cycles per instruction of 5. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 4 gigahertz. Assume that there are no stalls in the pipeline. The speed up achieved in this pipelined processor is __4__.

$t_n = 5 * \frac{1}{5GHz} = 1ns$

$t_p = \frac{1}{4GHz} = 0.25ns$

$S = \frac{t_n}{t_p}$

$= \frac{1}{0.25}$

$= 4$



2 mins Summary

Topic Pipeline Processing

Topic Pipeline Cycle Time

Topic Speed Up



Happy Learning

THANK - YOU